HW4

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Q1

Let $\underline{x}$ be an $n \times 1$ random vector with mean vector $\underline{\mu}$ and variance-covariance matrix Σ. Use Theorem 3.1 to show that $\underline{x}$ has a multivariate normal distribution if and only if, for every $n \times 1$ vector of constants $\underline{a}$, $\underline{a}'\underline{x}$ has a (univariate) normal distribution.

Answer

We utilize the following results:

- Theorem 3.1: If the mgf's of two random vectors exist and are equal for all $\underline{t}$ in an open rectangle that includes the origin, then their cdf's are identical.
- Definition (n-variate normal of rank k): With $\text{rank}(\Sigma) = k \geq 1$, $\underline{x} \sim N(\underline{\mu}, \Sigma)$ if the cdf of $\underline{x}$ is the same as that of $\underline{\mu} + \Gamma'\underline{z}$, where Γ is a $k \times n$ matrix with $\Sigma = \Gamma'\Gamma$ (such a Γ exists by Lemma 2.15) and $\underline{z} \sim N(\underline{0}_k, I_k)$ is k-variate standard normal.
- mgf of $N(\underline{\mu}, \Sigma)$ (Section 3.1): $M_{\underline{x}}(\underline{t}) = \exp\left(\underline{t}'\underline{\mu} + \frac{1}{2}\underline{t}'\Sigma\underline{t}\right)$ for all $\underline{t} \in \mathbb{R}^n$; in particular, $N(m, v)$ has mgf $\exp\left(tm + \frac{1}{2}t^2 v\right)$ for all $t \in \mathbb{R}$.
- Corollary to Lemma 2.4: $\text{Var}(L'\underline{x}) = L'\Sigma L$, where $\Sigma = \text{Var}(\underline{x})$.

Proof: Both directions rest on the following identity, which follows from the definition of the mgf: For any $\underline{a} \in \mathbb{R}^n$ and $t \in \mathbb{R}$,

$$M_{\underline{a}'\underline{x}}(t) = \text{E}\left[\exp(t\underline{a}'\underline{x})\right] = \text{E}\left[\exp\left((t\underline{a})'\underline{x}\right)\right] = M_{\underline{x}}(t\underline{a}). \qquad (*)$$

($\Rightarrow$) Suppose $\underline{x} \sim N(\underline{\mu}, \Sigma)$, and let $\underline{a} \in \mathbb{R}^n$ be given. By $(*)$ and the mgf of $N(\underline{\mu}, \Sigma)$, for all $t \in \mathbb{R}$,

$$M_{\underline{a}'\underline{x}}(t) = M_{\underline{x}}(t\underline{a}) = \exp\left(t\underline{a}'\underline{\mu} + \frac{1}{2}t^2\underline{a}'\Sigma\underline{a}\right),$$

which is the mgf of $N(\underline{a}'\underline{\mu}, \underline{a}'\Sigma\underline{a})$. By Theorem 3.1, $\underline{a}'\underline{x} \sim N(\underline{a}'\underline{\mu}, \underline{a}'\Sigma\underline{a})$. Since $\underline{a}$ was arbitrary, every $\underline{a}'\underline{x}$ is normal.

($\Leftarrow$) Suppose $\underline{a}'\underline{x}$ is normal for every $\underline{a} \in \mathbb{R}^n$, and let $\underline{t} \in \mathbb{R}^n$ be given. Since $\text{E}(\underline{t}'\underline{x}) = \underline{t}'\underline{\mu}$ and $\text{Var}(\underline{t}'\underline{x}) = \underline{t}'\Sigma\underline{t}$ (Corollary to Lemma 2.4), $\underline{t}'\underline{x} \sim N(\underline{t}'\underline{\mu}, \underline{t}'\Sigma\underline{t})$. Taking $\underline{a} = \underline{t}$ and $t = 1$ in $(*)$ then gives

$$M_{\underline{x}}(\underline{t}) = M_{\underline{t}'\underline{x}}(1) = \exp\left(\underline{t}'\underline{\mu} + \frac{1}{2}\underline{t}'\Sigma\underline{t}\right).$$

Since $\underline{t}$ was arbitrary, $M_{\underline{x}}$ exists on all of $\mathbb{R}^n$ and agrees there with the mgf of $\underline{\mu} + \Gamma'\underline{z} \sim N(\underline{\mu}, \Sigma)$, for Γ and $\underline{z}$ as in the definition. By Theorem 3.1, $\underline{x}$ has the same cdf as $\underline{\mu} + \Gamma'\underline{z}$; that is, $\underline{x} \sim N(\underline{\mu}, \Sigma)$.

Combining both directions, we conclude that $\underline{x}$ has a multivariate normal distribution if and only if $\underline{a}'\underline{x}$ has a (univariate) normal distribution for every $\underline{a} \in \mathbb{R}^n$.

Q2

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$. Show that

$$\text{cov}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = 2\,\text{tr}(A\Sigma B\Sigma) + 4\underline{\mu}'A\Sigma B\underline{\mu},$$

where A and B are symmetric matrices of constants. [Hint: For any random variables w and z, $\text{Var}(w+z) = \text{Var}(w) + \text{Var}(z) + 2\,\text{cov}(w, z)$.]

Answer

We utilize the following results:

- Variance of a quadratic form (from Section 3.2): If $\underline{x} \sim N(\underline{\mu}, \Sigma)$ and C is a symmetric matrix of constants, then $\text{Var}(\underline{x}'C\underline{x}) = 2\,\text{tr}\left[(C\Sigma)^2\right] + 4\underline{\mu}'C\Sigma C\underline{\mu}$.

Proof: Following the hint, write $w \equiv \underline{x}'A\underline{x}$ and $z \equiv \underline{x}'B\underline{x}$, such that $w + z = \underline{x}'(A + B)\underline{x}$ with $A + B$ symmetric.

Applying the Section 3.2 result with $C = A + B$, (and then with $C = A$, and then with $C = B$):

$$\begin{aligned}
\text{Var}(w + z) &= 2\,\text{tr}\left[(A\Sigma + B\Sigma)^2\right] + 4\underline{\mu}'(A + B)\Sigma(A + B)\underline{\mu} \\
&= \text{Var}(w) + \text{Var}(z) + 2\left[\text{tr}(A\Sigma B\Sigma) + \text{tr}(B\Sigma A\Sigma)\right] + 4\left(\underline{\mu}'A\Sigma B\underline{\mu} + \underline{\mu}'B\Sigma A\underline{\mu}\right) \\
&= \text{Var}(w) + \text{Var}(z) + 4\,\text{tr}(A\Sigma B\Sigma) + 8\underline{\mu}'A\Sigma B\underline{\mu},
\end{aligned}$$

Note, we simplify this expression by using:

- A, B, and Σ symmetric,
- $\underline{\mu}'B\Sigma A\underline{\mu} = (\underline{\mu}'B\Sigma A\underline{\mu})' = \underline{\mu}'A\Sigma B\underline{\mu}$, and
- Known properties of the trace, i.e., $\text{tr}(B\Sigma A\Sigma) = \text{tr}(A\Sigma B\Sigma)$.

Returning then to the variance formula noted in the hint, subtracting $\text{Var}(w)$ and $\text{Var}(z)$ from $\text{Var}(w + z)$ above leaves

$$\begin{aligned}
\text{cov}(w, z) &= \frac{1}{2}\left[\text{Var}(w + z) - \text{Var}(w) - \text{Var}(z)\right] \\
&= \frac{1}{2}\left[4\,\text{tr}(A\Sigma B\Sigma) + 8\underline{\mu}'A\Sigma B\underline{\mu}\right] \\
&= 2\,\text{tr}(A\Sigma B\Sigma) + 4\underline{\mu}'A\Sigma B\underline{\mu}.
\end{aligned}$$

And we conclude that

$$\text{cov}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = 2\,\text{tr}(A\Sigma B\Sigma) + 4\underline{\mu}'A\Sigma B\underline{\mu},$$

Q3

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is an $n \times 1$ vector and $\text{rank}(\Sigma) \geq 1$. Let

$$w_1 = \underline{b}_1'\underline{x} + \underline{x}'A_1\underline{x} \quad \text{and} \quad w_2 = \underline{b}_2'\underline{x} + \underline{x}'A_2\underline{x},$$

where $\underline{b}_1$ and $\underline{b}_2$ are vectors of constants and A_1 and A_2 are symmetric matrices of constants. Let

$$\underline{c}_1 = \underline{b}_1 + 2A_1\underline{\mu} \quad \text{and} \quad \underline{c}_2 = \underline{b}_2 + 2A_2\underline{\mu}.$$

We utilize the following results for parts (a)–(b):

- Lemma 2.4: if L and M are matrices of constants having the same number of rows as $\underline{x}$ has elements, then $\text{cov}(L'\underline{x}, M'\underline{x}) = L'\Sigma M$, where $\Sigma = \text{Var}(\underline{x})$.
- Corollary to Theorem 3.7: if $\underline{x} \sim N(\underline{\mu}, \Sigma)$ and B_1, B_2 are matrices of constants with n columns, then $B_1\underline{x}$ and $B_2\underline{x}$ are independent if and only if $B_1\Sigma B_2' = 0$.
- Definition of a generalized inverse: $A_iA_i^-A_i = A_i$.
- Continuous functions of independent random vectors are independent.

(a)

Find

$$\text{cov}\left(\begin{pmatrix}\underline{b}_1' \\ A_1\end{pmatrix}\underline{x}, \begin{pmatrix}\underline{b}_2' \\ A_2\end{pmatrix}\underline{x}\right).$$

Answer

Note: Each $\underline{b}_i$ is $n \times 1$ and each A_i is $n \times n$. For $i = 1, 2$, write $B_i \equiv \begin{pmatrix}\underline{b}_i' \\ A_i\end{pmatrix}$, an $(n+1) \times n$ matrix.

Since A_i is symmetric, $B_i' = (\underline{b}_i, A_i)$, an $n \times (n+1)$ matrix.

Via Lemma 2.4, with $L = B_1'$ and $M = B_2'$, we know

$$\text{cov}(B_1\underline{x}, B_2\underline{x}) = B_1\Sigma B_2' = \begin{pmatrix}\underline{b}_1' \\ A_1\end{pmatrix}\Sigma\begin{pmatrix}\underline{b}_2 & A_2\end{pmatrix}.$$

Multiplying out blockwise, then:

$$B_1\Sigma = \begin{pmatrix}\underline{b}_1'\Sigma \\ A_1\Sigma\end{pmatrix} \quad \Longrightarrow \quad B_1\Sigma B_2' = \begin{pmatrix}\underline{b}_1'\Sigma \\ A_1\Sigma\end{pmatrix}\begin{pmatrix}\underline{b}_2 & A_2\end{pmatrix} = \begin{pmatrix}\underline{b}_1'\Sigma\underline{b}_2 & \underline{b}_1'\Sigma A_2 \\ A_1\Sigma\underline{b}_2 & A_1\Sigma A_2\end{pmatrix},$$

which leads us to say

$$\text{cov}\left(\begin{pmatrix}\underline{b}_1' \\ A_1\end{pmatrix}\underline{x}, \begin{pmatrix}\underline{b}_2' \\ A_2\end{pmatrix}\underline{x}\right) = \begin{pmatrix}\underline{b}_1'\Sigma\underline{b}_2 & \underline{b}_1'\Sigma A_2 \\ A_1\Sigma\underline{b}_2 & A_1\Sigma A_2\end{pmatrix}. \qquad \text{(Q3 (a))}$$

(b)

Suppose that (i) $\underline{b}_1'\Sigma\underline{b}_2 = 0$, (ii) $\underline{b}_1'\Sigma A_2 = \underline{0}'$, (iii) $\underline{b}_2'\Sigma A_1 = \underline{0}'$, and (iv) $A_1\Sigma A_2 = 0$. Use (a) to show that w_1 and w_2 are independent. [Hint: Note that $w_i = \underline{b}_i'\underline{x} + (A_i\underline{x})'A_i^-(A_i\underline{x})$ for $i = 1, 2$.]

Answer

Proof: Let B_i be as in part (a), and let $\underline{u}_i \equiv B_i\underline{x} = \begin{pmatrix} \underline{b}_i'\underline{x} \\ A_i\underline{x} \end{pmatrix}$ for $i = 1, 2$. By part (a), $\text{cov}(\underline{u}_1, \underline{u}_2) = B_1\Sigma B_2'$ is the matrix in (Q3 (a)), which is 0 by assumptions (i)–(iv), i.e., the elements of (Q3 (a)) are all zero. (Specifically, for the lower-left block, $A_1\Sigma\underline{b}_2 = (\underline{b}_2'\Sigma A_1)' = \underline{0}_n$ by assumption (iii), as A_1 and Σ are symmetric.)

Since $\underline{u}_1$ and $\underline{u}_2$ are linear transformations of the same normal $\underline{x}$, the Corollary to Theorem 3.7 then gives that $\underline{u}_1$ and $\underline{u}_2$ are independent.

By the hint (which holds as $A_iA_i^-A_i = A_i$ and $A_i' = A_i$), $w_i = \underline{b}_i'\underline{x} + (A_i\underline{x})'A_i^-(A_i\underline{x})$ is a continuous function of $\underline{u}_i$ (and only $\underline{u}_i$), leading us to conclude that w_1 and w_2 (as they are continuous functions of independent random vectors) are themselves independent.

Q4

Take the model to be Model II and assume that $\underline{y}$ has a multivariate normal distribution.

We utilize the following results for parts (a)–(b):

- Corollary to Theorem 2.5 and the Note following Theorem 2.6: For any solution $\widehat{\underline{\beta}}$ to the normal equations, $X\widehat{\underline{\beta}} = P_X\underline{y}$, so $\underline{y} - X\widehat{\underline{\beta}} = (I_n - P_X)\underline{y}$.
- Lemmas 2.6–2.7 and property (4) of P_X: $I_n - P_X$ is symmetric and idempotent of rank $n - p^*$, and $(I_n - P_X)X = 0$.
- Corollary to Lemma 2.5: If F is idempotent, then $\mathrm{tr}(F) = \mathrm{rank}(F)$.
- Lemma 2.8: If $\mathrm{E}(\underline{x}) = \underline{\mu}$ and $\mathrm{Var}(\underline{x}) = \Sigma$, then $\mathrm{E}(\underline{x}'C\underline{x}) = \mathrm{tr}(C\Sigma) + \underline{\mu}'C\underline{\mu}$.
- Variance of a quadratic form (from Section 3.2): If $\underline{x} \sim N(\underline{\mu}, \Sigma)$ and C is a symmetric matrix of constants, then $\mathrm{Var}(\underline{x}'C\underline{x}) = 2\,\mathrm{tr}\left[(C\Sigma)^2\right] + 4\underline{\mu}'C\Sigma C\underline{\mu}$.
- Mean squared error: For an estimator t of σ^2, $\mathrm{MSE}(t) \equiv \mathrm{E}[(t - \sigma^2)^2] = \mathrm{Var}(t) + [\mathrm{E}(t) - \sigma^2]^2$.

Note: Under Model II with $\underline{y}$ multivariate normal, $\underline{y} \sim N(X\underline{\beta}, \sigma^2 I_n)$, where X is $n \times p$ with $p^* = \mathrm{rank}(X)$; as in Section 2.8, we take $n > p^*$, and write $\widetilde{\sigma}^2_k \equiv (1/k)\underline{y}'(\underline{y} - X\widehat{\underline{\beta}})$.

(a)

Obtain the expectation, the variance, and the mean squared error for any estimator of σ^2 having the form $(1/k)\underline{y}'(\underline{y} - X\widehat{\underline{\beta}})$, where k is a nonzero constant, and $\widehat{\underline{\beta}}$ is any solution to the normal equations.

Answer

We start by writing $\widetilde{\sigma}^2_k$ as a quadratic form in $\underline{y}$. Since $X\widehat{\underline{\beta}} = P_X\underline{y}$ for any solution $\widehat{\underline{\beta}}$ (Corollary to Theorem 2.5),

$$\widetilde{\sigma}^2_k = \frac{1}{k}\underline{y}'(\underline{y} - P_X\underline{y}) = \frac{1}{k}\underline{y}'(I_n - P_X)\underline{y} = \frac{\sigma^2}{k}u, \quad \text{where} \quad u \equiv \underline{y}'\left[\frac{1}{\sigma^2}(I_n - P_X)\right]\underline{y},$$

so $\widetilde{\sigma}^2_k$ does not depend on the choice of $\widehat{\underline{\beta}}$.

Next, we find the distribution of u via Theorem 3.16, with $\underline{x} = \underline{y}$, $\underline{\mu} = X\underline{\beta}$, $\Sigma = \sigma^2 I_n$ (of rank n), and $A = (1/\sigma^2)(I_n - P_X)$, which is symmetric (Lemma 2.6). Since

$$A\Sigma = \frac{1}{\sigma^2}(I_n - P_X)\sigma^2 I_n = I_n - P_X$$

is idempotent (via Lemma 2.6) of rank $n - p^*$ (via Lemma 2.7), Theorem 3.16 gives $u \sim \chi^2(n - p^*, \lambda)$, where, by property (4) of P_X,

$$\lambda = \frac{1}{2}(X\underline{\beta})'\left[\frac{1}{\sigma^2}(I_n - P_X)\right]X\underline{\beta} = \frac{1}{2\sigma^2}\underline{\beta}'X'(X - P_X X)\underline{\beta} = \frac{1}{2\sigma^2}\underline{\beta}'X'(X - X)\underline{\beta} = 0.$$

That is, $u \sim \chi^2(n - p^*)$, so $\mathrm{E}(u) = n - p^*$ and $\mathrm{Var}(u) = 2(n - p^*)$ (Section 3.3 with $\lambda = 0$). It then follows that

$$\mathrm{E}(\widetilde{\sigma}^2_k) = \frac{\sigma^2}{k}\,\mathrm{E}(u) = \frac{(n - p^*)\sigma^2}{k} \quad \text{and} \quad \mathrm{Var}(\widetilde{\sigma}^2_k) = \left(\frac{\sigma^2}{k}\right)^2 \mathrm{Var}(u) = \frac{2(n - p^*)\sigma^4}{k^2},$$

the latter expression established by the Corollary to Lemma 2.4 (with $L = \sigma^2/k$).

Finally, the bias of $\widetilde{\sigma}^2_k$ is

$$\mathrm{E}(\widetilde{\sigma}^2_k) - \sigma^2 = \frac{(n - p^*)\sigma^2}{k} - \frac{k\sigma^2}{k} = \frac{(n - p^* - k)\sigma^2}{k},$$

so, by the definition of the MSE,

$$\mathrm{MSE}(\widetilde{\sigma}^2_k) = \frac{2(n - p^*)\sigma^4}{k^2} + \frac{(n - p^* - k)^2\sigma^4}{k^2} = \frac{\sigma^4}{k^2}\left[2(n - p^*) + (n - p^* - k)^2\right]. \qquad \text{(Q4 (a))}$$

And this expression holds for every nonzero k and every solution $\underline{\widehat{\beta}}$ to the normal equations.

(b)

Of those estimators of σ^2 having the above form, which has the smallest mean squared error? (That is, determine the optimum value of k.)

Answer

By (Q4 (a)), $\mathrm{MSE}(\widetilde{\sigma}_k^2)$ is σ^4 times a function of k. Since $\sigma^4 > 0$ does not depend on k, the k that minimizes $\mathrm{MSE}(\widetilde{\sigma}_k^2)$ also minimizes $\mathrm{MSE}(\widetilde{\sigma}_k^2)/\sigma^4$.

We therefore work with $\mathrm{MSE}(\widetilde{\sigma}_k^2)/\sigma^4$ for simplicity. To that end, write $m \equiv n - p^*$ (≥ 1, as $n > p^*$) and $v \equiv 1/k$. As k ranges over the nonzero reals, so does v; hence minimizing over k is the same as minimizing over v.

Then, since $(m-k)/k = m/k - 1 = mv - 1$, (Q4 (a)) gives

$$\begin{aligned}\frac{\mathrm{MSE}(\widetilde{\sigma}_k^2)}{\sigma^4} &= \frac{2m + (m-k)^2}{k^2} = \frac{2m}{k^2} + \left(\frac{m-k}{k}\right)^2 \\ &= 2mv^2 + (mv-1)^2 = 2mv^2 + m^2v^2 - 2mv + 1 \\ &= m(m+2)v^2 - 2mv + 1.\end{aligned}$$

Completing the square, as $m(m+2)\left(v - \frac{1}{m+2}\right)^2 = m(m+2)v^2 - 2mv + \frac{m}{m+2}$,

$$\frac{\mathrm{MSE}(\widetilde{\sigma}_k^2)}{\sigma^4} = m(m+2)\left(v - \frac{1}{m+2}\right)^2 + 1 - \frac{m}{m+2} = m(m+2)\left(v - \frac{1}{m+2}\right)^2 + \frac{2}{m+2}.$$

Since $m(m+2) > 0$, the first term is nonnegative and equals zero only at $v = 1/(m+2)$, so $\mathrm{MSE}(\widetilde{\sigma}_k^2)/\sigma^4 \geq 2/(m+2)$, with equality only at $v = 1/(m+2)$. What's more, $1/(m+2) \neq 0$, so this v corresponds to the nonzero value $k = 1/v = m+2$. Hence the optimum value is $k = n - p^* + 2$; that is, of the estimators having the above form, the one with the smallest MSE is

$$\widetilde{\sigma}_{n-p^*+2}^2 = \frac{\underline{y}'(\underline{y} - X\underline{\widehat{\beta}})}{n - p^* + 2}.$$

Q5

Let $\underline{x} \sim N(\underline{\mu}, \Sigma)$, where $\underline{x}$ is an $r \times 1$ vector and $\text{rank}(\Sigma) = k \geq 1$ (where possibly $k < r$). Let A be an $r \times r$ symmetric matrix. Show that, if $A\Sigma$ is idempotent of rank s and if in addition $\underline{\mu}$ lies in $\mathcal{C}(\Sigma)$, then $\underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right)$. (Hint: Let Γ be a $k \times r$ matrix such that $\Sigma = \Gamma'\Gamma$, and show that $A\Sigma$ is idempotent of rank $s \implies \Gamma A\Gamma'$ is idempotent of rank s. Then, use $\underline{x}$ has the same distribution as $\Gamma'(\underline{z} + \underline{a})$, where $\underline{z}$ is a standard normal vector and $\underline{a}$ is a certain vector of constants.)

Answer

We utilize the following results:

- Definition of $N(\underline{\mu}, \Sigma)$: With $\text{rank}(\Sigma) = k \geq 1$, $\underline{x} \sim N(\underline{\mu}, \Sigma)$ if the cdf of $\underline{x}$ is the same as that of $\underline{\mu} + \Gamma'\underline{z}$, where Γ is a $k \times r$ matrix with $\Sigma = \Gamma'\Gamma$ (such a Γ exists by Lemma 2.15) and $\underline{z} \sim N(\underline{0}_k, I_k)$ is k-variate standard normal.
- Lemma 1.3(i): For any matrix X, $PX'X = QX'X$ implies $PX' = QX'$.
- Corollary to Lemma 2.5: If F is idempotent, then $\text{tr}(F) = \text{rank}(F)$.
- Theorem 3.5: If $\underline{x} \sim N(\underline{\mu}, \Sigma)$, then $B\underline{x} + \underline{b} \sim N(B\underline{\mu} + \underline{b}, B\Sigma B')$.
- Theorem 3.16 (with $\Sigma = I$): If $\underline{v} \sim N(\underline{a}, I_k)$ and M is a $k \times k$ symmetric matrix of constants that is idempotent of rank $s \geq 1$, then $\underline{v}'M\underline{v} \sim \chi^2\left(s, \frac{1}{2}\underline{a}'M\underline{a}\right)$.

Note: We take $s \geq 1$, as in Theorem 3.16.

Proof: As in the hint, let Γ be a $k \times r$ matrix with $\Sigma = \Gamma'\Gamma$ (such a Γ exists by Lemma 2.15). We start by showing that $\Gamma A\Gamma'$ is idempotent of rank s. Since $A\Sigma$ is idempotent,

$$(A\Sigma A)\Gamma'\Gamma = (A\Sigma)^2 = A\Sigma = A\Gamma'\Gamma,$$

so Lemma 1.3(i), with Γ in the role of X, gives $A\Sigma A\Gamma' = A\Gamma'$.

It then follows that

$$(\Gamma A\Gamma')^2 = \Gamma A(\Gamma'\Gamma)A\Gamma' = \Gamma(A\Sigma A\Gamma') = \Gamma A\Gamma',$$

so the matrix $\Gamma A\Gamma'$ is symmetric (as A is symmetric) as well as idempotent.

Then, by the Corollary to Lemma 2.5 (twice applied!) and using the known property of the trace, the rank of $\Gamma A\Gamma'$ is

$$\text{rank}(\Gamma A\Gamma') = \text{tr}(\Gamma A\Gamma') = \text{tr}(A\Gamma'\Gamma) = \text{tr}(A\Sigma) = \text{rank}(A\Sigma) = s.$$

Next, we identify the vector $\underline{a}$ in the hint. Since $\underline{\mu} \in \mathcal{C}(\Sigma)$, $\underline{\mu} = \Sigma\underline{d}$ for some $\underline{d} \in \mathbb{R}^r$. It then follows that $\underline{\mu} = \Gamma'\Gamma\underline{d} = \Gamma'\underline{a}$, where $\underline{a} \equiv \Gamma\underline{d} \in \mathbb{R}^k$.

By the definition, $\underline{x}$ then has the same cdf as

$$\underline{\mu} + \Gamma'\underline{z} = \Gamma'(\underline{z} + \underline{a}) = \Gamma'\underline{v},$$

where $\underline{z} \sim N(\underline{0}_k, I_k)$ and $\underline{v} \equiv \underline{z} + \underline{a} \sim N(\underline{a}, I_k)$ (via Theorem 3.5). Hence $\underline{x}'A\underline{x}$ has the same distribution as $(\Gamma'\underline{v})'A(\Gamma'\underline{v}) = \underline{v}'(\Gamma A\Gamma')\underline{v}$.

Taken together, since $\Gamma A\Gamma'$ is symmetric and idempotent of rank s, Theorem 3.16 gives $\underline{v}'(\Gamma A\Gamma')\underline{v} \sim \chi^2\left(s, \frac{1}{2}\underline{a}'\Gamma A\Gamma'\underline{a}\right)$. And, as $\underline{a}'\Gamma A\Gamma'\underline{a} = (\Gamma'\underline{a})'A(\Gamma'\underline{a}) = \underline{\mu}'A\underline{\mu}$, we may then conclude

$$\underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right).$$

Q6

Let $\underline{x} \sim N(\underline{0}, I)$, and take A and B to be two symmetric matrices such that $\underline{x}'A\underline{x} \sim \chi^2(p)$ and $\underline{x}'B\underline{x} \sim \chi^2(q)$, where $p > q$. Show, by completing the following three steps, that $\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q)$ if and only if the correlation $\text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = q/\sqrt{pq}$.

We utilize the following results for parts (a)–(c):

- Q2, which here ($\underline{\mu} = \underline{0}_n$, $\Sigma = I_n$) reads $\text{cov}(\underline{x}'C\underline{x}, \underline{x}'D\underline{x}) = 2\,\text{tr}(CD)$ for symmetric C and D.
- Section 3.3 (stated in Q4): If $u \sim \chi^2(m, \lambda)$, then $\text{Var}(u) = 2(m + 4\lambda)$.
- Theorem 3.16 (stated in Q4) and its converse: If $\underline{x}'M\underline{x} \sim \chi^2(s, \lambda)$ for some λ, then $M\Sigma$ is idempotent of rank s.
- Corollary to Lemma 2.5: If F is idempotent, then $\text{tr}(F) = \text{rank}(F)$.
- Lemma 1.2: For any matrix X, $\text{tr}(X'X) = 0$ implies $X = 0$.
- Lemma 3.6: If F, G, and $F + G$ are all idempotent, then $FG = 0$.

Note: Let n denote the length of $\underline{x}$. By the converse in Theorem 3.16 (with $\Sigma = I_n$), A and B are idempotent of ranks p and q, resp., so $\text{tr}(A) = p$ and $\text{tr}(B) = q$ (Corollary to Lemma 2.5), $I_n - A$ is symmetric and idempotent, and $p - q \geq 1$ (as $p > q$ are integers).

(a)

Show that $\text{tr}[(I-A)B] = 0$ if and only if $\text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = q/\sqrt{pq}$. [Hint: Consider $\text{cov}(\underline{x}'(A-B)\underline{x} + \underline{x}'B\underline{x}, \underline{x}'B\underline{x})$]

Answer

Proof: Following the hint, by linearity of the covariance,

$$\begin{aligned}\text{cov}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) &= \text{cov}(\underline{x}'(A-B)\underline{x} + \underline{x}'B\underline{x}, \underline{x}'B\underline{x}) \\ &= \text{cov}(\underline{x}'(A-B)\underline{x}, \underline{x}'B\underline{x}) + \text{Var}(\underline{x}'B\underline{x}).\end{aligned}$$

By Q2 (as $A - B$ and B are symmetric) and $B^2 = B$, the first term is $2\,\text{tr}[(A-B)B] = 2\,\text{tr}(AB - B) = -2\,\text{tr}[(I_n - A)B]$, while $\text{Var}(\underline{x}'A\underline{x}) = 2p$ and $\text{Var}(\underline{x}'B\underline{x}) = 2q$ (Section 3.3 with $\lambda = 0$). Hence $\text{cov}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = 2q - 2\,\text{tr}[(I_n - A)B]$, and

$$\text{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = \frac{2q - 2\,\text{tr}[(I_n - A)B]}{\sqrt{2p \cdot 2q}} = \frac{q - \text{tr}[(I_n - A)B]}{\sqrt{pq}},$$

which equals $q/\sqrt{pq}$ if and only if $\text{tr}[(I_n - A)B] = 0$.

(b)

Show that $(I-A)B = 0$ if and only if $\text{tr}[(I-A)B] = 0$.

Answer

Proof: ($\Rightarrow$) If $(I_n - A)B = 0$, then $\text{tr}[(I_n - A)B] = \text{tr}(0) = 0$.

($\Leftarrow$) Write $M \equiv B(I_n - A)$, so that $M' = (I_n - A)B$. Since $(I_n - A)^2 = I_n - A$ and $B^2 = B$, the cyclic property of the trace gives

$$\text{tr}[(I_n - A)B] = \text{tr}[(I_n - A)(I_n - A)BB] = \text{tr}[(I_n - A)BB(I_n - A)] = \text{tr}(M'M).$$

So if $\text{tr}[(I_n - A)B] = 0$, then Lemma 1.2 gives $M = 0$, and hence $(I_n - A)B = M' = 0$.

Combining both directions, $(I_n - A)B = 0$ if and only if $\text{tr}[(I_n - A)B] = 0$.

(c)

Show that $\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q)$ if and only if $(I-A)B = 0$.

Answer

Proof: Since $A-B$ is symmetric and $p-q \geq 1$, Theorem 3.16 and its converse give $\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q)$ if and only if $A-B$ is idempotent of rank $p-q$ (the noncentrality is 0, as $\underline{\mu} = \underline{0}_n$). It then suffices to show that $A-B$ is idempotent of rank $p-q$ if and only if $(I_n - A)B = 0$.

($\Rightarrow$) If $A-B$ is idempotent, then $A-B$, B, and $(A-B)+B = A$ are all idempotent, so Lemma 3.6 gives $(A-B)B = AB - B = 0$; that is, $(I_n - A)B = 0$.

($\Leftarrow$) If $(I_n - A)B = 0$, then $AB = B$, and transposing, $BA = B$. Hence

$$(A-B)^2 = A^2 - AB - BA + B^2 = A - B - B + B = A - B,$$

so $A-B$ is idempotent, of rank $\mathrm{tr}(A-B) = \mathrm{tr}(A) - \mathrm{tr}(B) = p-q$ (Corollary to Lemma 2.5).

Combining both directions, $\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q)$ if and only if $(I_n - A)B = 0$.

Then, taking parts (a)–(c) together, we may then conclude that

$$\underline{x}'(A-B)\underline{x} \sim \chi^2(p-q) \overset{\text{(c)}}{\iff} (I_n - A)B = 0 \overset{\text{(b)}}{\iff} \mathrm{tr}[(I_n - A)B] = 0 \overset{\text{(a)}}{\iff} \mathrm{corr}(\underline{x}'A\underline{x}, \underline{x}'B\underline{x}) = \frac{q}{\sqrt{pq}}.$$